

WEBSITE: PIXIU FLOW

Profit Tracking Website for Micro-Small Businesses Group Footprint Report

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Profit Tracking Website for Micro-Small Businesses
Group Footprint Report

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Product Name	Profit Tracking Website for Micro-Small Businesses
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I. Problem Our Group Wants to Solve

Pixiu Flow aims to solve the problem of weak financial visibility among micro-enterprises and household businesses in Vietnam. Many very small businesses record sales and expenses manually through notebooks, phone notes, memory or do not record at all. Using other services on the market such as KiotViet or simply using Excel to record require some degree of financial knowledge and expertise to navigate complicated interfaces.

This makes it difficult for owners to know their actual revenue, expenses, profit margins, inventory status, and unpaid customer bills. The issue is not only that financial data is recorded inefficiently, but that important business information remains scattered and disconnected.

This problem occurs in the context of Vietnam's large micro-business sector. Micro-enterprises and household businesses form a major part of the Vietnamese economy, but many operate with limited staff, limited accounting knowledge, and limited access to affordable digital management tools. In the formal enterprise sector, OECD data based on Vietnam's 2016 General Statistics Office Enterprise Census shows that 70% of all formal firms were micro-enterprises with fewer than 10 employees. Beyond formal firms, Vietnam also has millions of household businesses. The Ministry of Planning and Investment reported that Vietnam had over 5 million business households in 2024, while the Ministry of Finance reported around 5.2 million active household businesses by the end of 2023. These businesses are usually small in scale, but collectively represent an important source of employment, income generation, and local economic activity. Many of these businesses are owned by those unfamiliar with technological devices or financial calculations.

The affected users are mainly micro-business owners, household business owners, online sellers, small shop owners, student entrepreneurs, and small service providers. For these users, financial management problems can quickly become business survival problems. If revenue and expenses are not properly categorised, owners may not know which products or services are actually profitable. If inventory is not tracked, capital may be locked in slow-moving stock, or the business may run out of best-selling items. If unpaid customer bills are not monitored, the business may report sales but still face cash shortages.

Pixiu Flow addresses this problem by helping users track revenue, expenses, inventory, margins, financial ratios, and unpaid customer bills in one simple platform, in a user-friendly, simple interface that does not require any previous financial training. The goal is to give very small businesses a clearer view of their financial position. By turning scattered records into organized business insights, Pixiu Flow helps users understand not only how much they sell, but how much they actually earn, where money is being spent, which categories are profitable, and which customers still need to pay.

II. Target Users

The main target users of Pixiu Flow are micro-business owners, household business owners, online sellers, small shop owners, student entrepreneurs, and small service providers in Vietnam. These users usually operate with limited staff and limited financial-management knowledge. In many cases, the owner is responsible for multiple tasks at the same time, including selling products, buying inventory, recording expenses, collecting customer payments, and checking whether the business is profitable.

These users need a tool that is simple enough to use daily but structured enough to provide meaningful business information. They do not only need a place to record transactions; they need to understand what those transactions mean. Specifically, they need to know how much revenue they generate, how much they spend, which product or category creates the highest profit, how much inventory remains, which items are slow-moving, and which customers have not paid their bills. For very small businesses, these details are important because cash flow, inventory, and profit margins directly affect survival.

Pixiu Flow is designed for practical business situations. A user may use the product after completing a sale to record revenue, after purchasing new stock to update inventory, or at the end of the day to review business performance. The product can also be used weekly or monthly to check profit margins, compare revenue and expenses across categories, identify unpaid customer bills, and decide whether to restock, reduce costs, or adjust selling prices. In this way, Pixiu Flow supports both daily transaction recording and broader business decision-making.



III. What the Product Can Currently Do

At the current stage, Pixiu Flow has been designed as a revenue, expense, profit, and inventory tracking platform for micro and small businesses.

The product allows users to create an account, log in, and set up a business profile. During setup, users can enter basic business information such as business name, business type, country, currency, existing revenue and expense categories. This helps the system organize data according to the user's actual business model.

The product also supports transaction recording. Users can log new sales transactions by entering information such as product or service category, price, quantity, and customer. They can also record business expenses, including inventory purchases and other operating expenses. For users with many transactions, the product includes an Excel-like multiple-transaction input format, making data entry faster and more familiar for small business owners.

Pixiu Flow can track revenue and expenses by category. This means users can see not only total revenue and total cost, but also which categories contribute most to sales and which categories create the largest expenses. Based on these inputs, the system calculates key financial indicators such as gross profit, net profit, profit margin, category-level sales, and category-level profitability. These outputs are shown through dashboard summary cards, charts, and breakdown sections.

The product also includes inventory tracking. When users record inventory purchases, the system updates stock quantity and inventory value. When users record sales, inventory can be reduced accordingly. The inventory section shows product lines, stock quantity, inventory value, days in inventory, best-selling items, slow-moving items, and items with strong or weak profit margins. This helps users avoid overstocking products that do not sell well and running out of products that sell quickly.

Another important function is customer payment tracking. Pixiu Flow is designed to record unpaid customer bills and remind users after a pre-set time period. This feature is especially useful for micro-businesses that allow customers to pay later, because unpaid bills can create cash flow problems even when sales appear strong.

Overall, the current product logic allows users to move from scattered manual records to a more organized business-management process. Pixiu Flow can help users record transactions, revenue and expenses, monitor inventory, calculate margins and financial ratios, track unpaid bills, and view business insights in one platform. Instead of only showing whether the business has sales, the product helps users understand whether the business is actually profitable and financially manageable.

IV. Input

1. Landing Page	
Element / Feature	Description
Login Button	Navigation action labelled “Đăng nhập” that allows existing users to move from the landing page to the login page. This is an entry input for returning users.
Sign Up Button	Navigation action labelled “Đăng kí” that allows new users to start the registration and onboarding process.
Primary CTA Button	Main call-to-action button labelled “Dùng thử miễn phí” that leads new users toward the sign-up or product trial flow.
Secondary CTA Button	Secondary call-to-action button labelled “Xem demo” with a play icon. This allows visitors to view or access the product demonstration before creating an account.
Business Type Selection Cards	Informational category cards such as retail shop, café/restaurant, online shop, and personal service. These do not collect typed data, but they communicate the target user groups and guide users to understand whether the product fits their business type.
Feature Exploration Sections	Sections such as feature highlights and “About Us” allow visitors to read product information before deciding whether to register. These are not financial inputs, but they affect the user’s decision to continue into the system.

2. Login Page	
Element / Feature	Description
Username / Email / Phone Field	Text input labelled “Tên đăng nhập / Email / Số điện thoại.” Users enter their username, email address, or phone number to identify their account. .
Password Field	Password input labelled “Mật khẩu.” Users enter their password to authenticate access to PIXIU FLOW. The entered characters are hidden for privacy.
Continue with Google Button	Alternative login action labelled “Tiếp tục với Google.” Users can authenticate through Google instead of entering manual login credentials.
Register Now Link	Link labelled “Đăng ký ngay” for users who do not have an account. This redirects them to the sign-up and onboarding flow.

3. Sign Up / Onboarding Flow

Element / Feature	Description
Email / Phone Field	Text input used to enter the email address or phone number for the new admin account. This information is used for account creation and later login.
Business Owner Name	Text input labelled 'Tên chủ doanh nghiệp.' Users enter the name of the business owner or main account holder.
Business / Store Name	Text input labelled 'Tên doanh nghiệp / Cửa hàng.' Users enter the name of the business or store that will be managed in PIXIU FLOW.
Business Type	Dropdown labelled 'Loại hình kinh doanh.' Users select the business type, such as retail, cafe/restaurant, online shop, service business, or another suitable category.
Main Currency	Dropdown labelled 'Đơn vị tiền tệ chính.' Users select the main currency used by the business, such as VND - Việt Nam Đồng. This affects how financial values are displayed.
Country Field	Input or dropdown labelled 'Quốc gia.' Users select the country where the business operates, such as Việt Nam.
Product Category	Onboarding step for setting up product categories. Users can define the product groups used later in inventory and sales transactions.
Expense Category	Onboarding step for setting up expense categories. Users can define the cost groups used later when recording business expenses.

4. Initial Inventory Declaration Page

Element / Feature	Description
Initial Inventory Table	Table used to declare goods that already existed before the business started using PIXIU FLOW. This input sets the opening inventory balance and is not treated as a new purchase transaction.
Product Name Field	Required text input labelled 'Tên sản phẩm.' Users enter the name of an existing product, such as coffee, milk, instant noodles, or other items already available in stock.
Product Category	Required dropdown labelled 'Danh mục.' Users select the product category for each item. This connects the product to inventory grouping and later reporting.
Unit Field	Input labelled 'Đơn vị.' Users enter or select the unit of measurement, such as pack, box, bottle, piece, kilogram, or other units.

Selling Price	Required numeric input labelled 'Giá bán.' Users enter the expected selling price of the product. This value can later support sales entry and profit estimation.
Estimated Cost Price	Numeric input labelled 'Giá vốn ước tính.' Users enter the estimated cost price of the product already in stock. This is used to calculate opening inventory value.
Initial Stock Quantity	Required numeric input labelled 'Tồn kho ban đầu.' Users enter the quantity already available before using the software. This becomes the opening stock quantity.
Stock Date	Date input labelled 'Ngày nhập kho.' Users can enter an estimated date related to the existing stock record. This helps support inventory tracking, but it does not create cash outflow.
Low-Stock Alert Threshold	Numeric input labelled 'Ngưỡng cảnh báo.' Users enter the minimum acceptable stock level. If current stock is lower than or equal to this value, the system marks the product as low stock.

5. Transaction Record Page - Revenue

Element / Feature	Description
Customer / Partner	Optional dropdown or search field labelled 'Khách hàng.' Users select or enter the customer or partner related to the sale.
Transaction Date	Date picker labelled 'Ngày giao dịch.' Users enter the date of the revenue transaction.
Product Name	Product selection input inside the order detail table. Users select or enter the product being sold.
Unit Selling Price	Numeric input labelled 'Đơn giá.' Users enter the selling price per unit.
Quantity	Numeric input labelled 'Số lượng.' Users enter the number of units sold.
Payment Method	Dropdown labelled 'Phương thức thanh toán.' Users choose how the customer paid, such as cash or another payment method.
Receivable Reminder Date	Date field labelled 'Ngày nhắc thu nợ.' Users can enter a reminder date if the customer has not paid immediately or if follow-up is needed.
Invoice / Note	Text area labelled 'Ghi chú / Hóa đơn.' Users can add notes, invoice details, or supporting information for the transaction.

6. Transaction Record Page - Purchase Expense

Element / Feature	Description
Transaction Date	Date picker labelled 'Ngày giao dịch.' Users enter the date when the cost occurs.
Supplier	Input labelled 'Nhà cung cấp.' Users enter or select the supplier for inventory purchase costs.
Purchased Product	Dropdown or product selector labelled 'Sản phẩm.' Users select the product being purchased.
Purchase Unit Price	Numeric input labelled 'Đơn giá nhập.' Users enter the purchase price per unit.
Purchase Quantity	Numeric input labelled 'Số lượng.' Users enter the quantity purchased.
Payment Method	Dropdown labelled 'Phương thức thanh toán.' Users choose how the expense or purchase is paid.
Payment Due Date	Date input labelled 'Ngày cần thanh toán.' Users enter a due date if the expense has not been paid immediately.
Purchase Note	Text area labelled 'Ghi chú nhập hàng.' Users add internal notes about the purchase.

7. Transaction Record Page - Other Expense

Element / Feature	Description
Transaction Date	Date picker labelled 'Ngày giao dịch'. The user enters the date when the other expense occurs.
Other Expense Detail Table	Dashed-border table labelled 'Chi tiết chi phí'. It contains one or more expense lines and shows the number of items currently being entered.
Expense Group	Dropdown labelled 'Nhóm chi phí'. The user selects the expense category, such as electricity, rent, delivery, packaging, marketing, salary, utilities, or another operating cost group.
Amount	Numeric input labelled 'Số tiền (VND)'. The user enters the amount of the selected expense line.

Expense	Text input labelled 'Ghi chú'. The user can add a short explanation or supporting note for the expense line.
Payment Method	Dropdown labelled 'Phương thức thanh toán'. The user selects how the expense is paid, such as cash or another payment method.
Payment Due Date	Date field labelled 'Ngày cần thanh toán'. The user enters a due date when the expense has not been paid immediately or needs later follow-up.

8. Multiple Transaction Entry Page

Element / Feature	Description
Add Row Button	Button labelled 'Thêm dòng.' Users click it to add a new transaction row manually.
Row Selection Checkbox	Checkbox at the beginning of each row. Users select rows that they want to delete or manage together.
Transaction Type	Dropdown in each row, such as revenue or cost. Users select the type of transaction for that row.
Date Field	Date input in each row. Users enter the transaction date.
Product / Cost Group	Dropdown or selector for choosing the product, product group, or cost group related to the transaction.
Add Product	Button labelled 'Thêm sản phẩm.' Users can add product details under a transaction row when needed.
Customer / Supplier	Text input for customer, partner, or supplier name. This identifies the external party related to the transaction.
Value	Numeric input for the transaction value or unit value, depending on the row structure.
Quantity	Numeric input for the quantity involved in the transaction.
Payment Method	Dropdown for selecting the payment method, such as cash or another method.
Due Date	Optional date field for payment deadline or debt tracking.
Note	Text input for additional transaction notes.

9. Inventory management

Element / Feature	Description
Product Name	Required text input labelled 'Tên sản phẩm.' Users enter the product name, such as instant noodles, milk, or coffee.
Unit of Measurement	Text input labelled 'Đơn vị tính.' Users enter the unit used to count the product, such as piece, box, pack, bottle, or kilogram.
Expected Selling Price	Numeric input labelled 'Giá bán dự kiến.' Users enter the expected selling price of the product.
Stock Alert Threshold	Numeric input labelled 'Cảnh báo tồn kho.' Users enter the minimum stock level used for low-stock warnings.
Existing Category	Dropdown labelled 'Danh mục cấp 1.' Users select an existing category for the product.
New Category	Text input labelled 'Hoặc thêm danh mục cấp 1.' Users type a new category name if the product does not fit existing categories.
Add Category	Plus button beside the new category field. Users click it to create the new category.
Create Product Only Option	Radio option labelled 'Đề sau.' Users select this if they only want to create the product profile now and add stock later.
Stock In Now Option	Radio option labelled 'Nhập kho ngay.' Users select this if they want to create the product and record stock quantity at the same time.
Stock-In Quantity	Numeric input labelled 'Số lượng nhập.' Users enter the quantity being added to inventory.
Cost Price per Unit	Numeric input labelled 'Giá vốn / đơn vị.' Users enter the cost price per unit for the stock-in record.
Supplier	Text input labelled 'Nhà cung cấp.' Users enter the supplier related to the stock-in transaction.
Stock-In Date	Date input labelled 'Ngày nhập hàng.' Users enter the date of the stock-in transaction.
Stock-In Note	Text area labelled 'Ghi chú.' Users add notes about the stock-in record.
Payment Method	Dropdown labelled 'Phương thức thanh toán.' Users select how the stock-in transaction is paid.

V. Logic or Processing Rules

1. Landing Page

The landing page acts as the first entry point of PIXIU FLOW. It does not process financial or inventory data yet. Its main purpose is to introduce the product, explain the value of the website, and guide visitors to the next action.

When a visitor clicks Login, the system redirects them to the login page. When a visitor clicks Sign Up or Try for Free, the system moves them to the registration and onboarding flow. If the visitor selects Watch Demo, the system should show a preview or demonstration of how PIXIU FLOW works.

The landing page also presents the target users, such as retail shops, cafés, restaurants, online shops, and personal service businesses. These sections do not change system data, but they help users understand whether the product is suitable for their business.

2. Login Page

The login page is used to verify returning users before allowing them to access the main system. Users can enter their username, email address, or phone number together with their password.

After the user clicks Login, PIXIU FLOW checks whether the account information is valid. If the login details are correct, the system redirects the user to the dashboard. If the information is incorrect, the system keeps the user on the login page and asks them to check the username, email, phone number, or password again.

The Remember Login option allows the system to keep the user signed in or make future login faster. The Show/Hide Password icon helps users check their password input before submitting. The Forgot Password link allows users to start the account recovery process if they cannot remember their password. If users choose Continue with Google, the system uses Google authentication instead of manual username and password entry.

3. Sign Up and Onboarding Flow

The sign-up process creates the user's admin account and business profile. PIXIU FLOW first asks for account information, including email or phone number, business owner's name, password, and password confirmation. The password confirmation field helps reduce password setup mistakes. If the password and confirmation do not match, the system should ask the user to correct the information before continuing.

After account creation, the system moves to the business profile setup step. Users enter the business or store name, business type, main currency, and country. These details are important because they become the foundation for later financial tracking and reporting. The selected currency controls how financial values are displayed. For example, if the user selects VND, revenue, cost, cash paid, profit, and inventory value will be shown in Vietnamese Dong. The business type helps the system understand the user's business context and prepare relevant features such as product management, inventory tracking, and transaction recording.

If the onboarding flow includes product categories and expense categories, the system stores these categories for later use. Product categories help organize inventory and sales records, while expense categories help classify business costs. Once onboarding is completed, the system redirects the user to the dashboard or main workspace

4. Initial Inventory Declaration

The initial inventory declaration page is used to record goods that the business already had before starting to use PIXIU FLOW. This step is different from recording a new purchase.

When users enter product name, category, unit, selling price, estimated cost price, initial stock quantity, stock date, and low-stock alert threshold, the system creates product records and opening inventory balances. The quantity entered in this page becomes the product's opening stock. This opening quantity is used as the starting point for future inventory updates. The main inventory formula is: $\text{Current Stock} = \text{Opening Stock} + \text{Stock In} - \text{Stock Out}$. The system also calculates the beginning inventory value by using the estimated cost price: $\text{Initial Inventory Value} = \text{Initial Stock Quantity} \times \text{Estimated Cost Price}$. A key rule is that initial inventory declaration does not create cash paid and does not affect cash flow. This is because the products were already owned by the business before the user started using PIXIU FLOW. Therefore, this action only updates inventory quantity and inventory value. It should not be treated as a new expense or purchase.

If users have just bought new goods after starting to use PIXIU FLOW, they should use the Stock-In or Inventory Purchase function instead. That type of transaction can increase stock and affect cash paid.

If users only want to create a product profile without stock quantity, they can choose Add Product Only. In that case, the system saves the product information but does not increase inventory.

Before saving the initial inventory table, the system should check whether required fields are completed. Product name, category, unit, cost price, selling price, initial quantity, and alert threshold should be valid. If there are missing or incorrect values, the user should correct them before saving.

After the initial stock is saved, the inventory page and inventory-related indicators are updated. However, revenue, expenses, cash paid, and net cash flow remain unchanged until real transactions are recorded.

5. Dashboard

The dashboard summarizes business performance based on the data recorded in the system. It displays key indicators such as recorded revenue, gross profit, net profit, cash received, cash paid, and net cash flow.

If the business has not entered any transaction or inventory data yet, the dashboard shows empty values or placeholder symbols. In this situation, PIXIU FLOW displays a getting-started guide. The guide suggests three actions: declare existing stock, record the first transaction, and view more indicators.

The date filter controls what data is included in the dashboard. Users can select options such as All, Today, 7 Days, This Month, This Year, or a custom date range. After the user clicks Apply, the system recalculates dashboard figures using only the transactions within the selected period. Recorded revenue is calculated from sales transactions. Gross profit is calculated by subtracting cost of goods sold from revenue. Net profit is calculated after deducting other operating costs. The dashboard also calculates cash movement. In PIXIU FLOW, cash paid refers to money paid for inventory purchases or other business expenses. Net cash flow compares revenue with cash paid: $\text{Net Cash Flow} = \text{Revenue} - \text{Cash Paid}$. This means cash flow focuses on money movement, while profit focuses on business performance after costs are considered.

The trend chart and quick statistics section use the same filtered data. When the user changes the date range, the chart and statistics are refreshed. This helps users review business results by day, week, month, year, or a selected period.

6. Revenue Transaction Recording

The revenue transaction page is used to record sales. When the user selects New Revenue, the system displays a form for entering customer, transaction date, product details, selling price, quantity, payment method, and notes.

For each product line, the system calculates revenue by multiplying unit selling price by quantity sold: $\text{Revenue} = \text{Selling Price} \times \text{Quantity Sold}$. If one transaction contains several products, the system adds all product line totals together to calculate total revenue. After the user saves the transaction, the system updates several parts of the website. The sales amount is added to recorded revenue. The transaction is stored in transaction history. The dashboard is recalculated. If the product is connected to inventory, the sold quantity is deducted from current stock. For example, if the business sells 50 packs of instant noodles at 6,000 VND per pack, the system calculates: $\text{Revenue} = 50 \times 6,000 = 300,000 \text{ VND}$

The system should also check whether the required transaction information has been entered correctly. At minimum, a revenue transaction should include transaction date, product, selling price, quantity, and payment method. If important fields are missing, the transaction should not be saved until the user corrects the form. If users add more product lines, the system recalculates the total revenue immediately. If users delete a product line, the total amount is reduced accordingly.

7. Expense Transaction Recording - Inventory Purchase

When the user selects New Cost, PIXIU FLOW asks whether the cost is an inventory purchase or another expense. If the user chooses Inventory Purchase Cost, the system displays product purchase fields.

For inventory purchases, users enter supplier, transaction date, purchased product, purchase unit price, quantity, payment method, due date if needed, and notes. The system calculates the purchase amount as: $\text{Purchase Cost} = \text{Purchase Unit Price} \times \text{Purchase Quantity}$

After the user saves the inventory purchase, the system increases the stock quantity of the selected product. The purchase also updates inventory value and appears in transaction history. If the purchase is paid immediately, the amount is included in cash paid. If the payment is due later, the system can record the due date so that the business owner can track future payment obligations. Inventory purchase affects stock and cash paid, but it is not the same as cost of goods sold. Cost of goods sold relates to products that have already been sold. Inventory purchase records money spent to buy goods, even if those goods have not been sold yet. This distinction is important because buying inventory can reduce cash flow immediately, while profit is only affected when the goods are sold or when expenses are recognized according to the system's calculation method.

8. Expense Transaction Recording - Other Expense

The Other Expense form is used to record business costs that are not inventory purchases. Examples include electricity, rent, delivery fees, packaging, marketing, wages, internet fees, or other operating expenses. When users select Other Expense, the form changes from product purchase details to expense details. Users enter the expense group, amount, note, payment method, and payment due date if needed. The system adds each expense line to the total cost of the transaction. If users add more expense rows, the total cost increases. If users delete an expense row, the total is reduced.

Other expenses do not change inventory quantity because they are not related to buying products for resale. However, they affect business cost, cash paid, and net profit. If the expense is paid immediately, it is included in cash paid. If it is not paid yet, the due date helps the user monitor future payment requirements.

After the user clicks Save Cost, the system stores the expense in transaction history and refreshes dashboard indicators. Net profit decreases because operating costs increase. Net cash flow also decreases if the expense is counted as cash paid.

The system should validate the expense before saving. The expense should include at least transaction date, expense group, amount, and payment method. If the amount is missing or invalid, the user should correct it before saving.

9. Multiple Transaction Entry

The multiple transaction entry page allows users to record several transactions at the same time. Each row represents one transaction or one transaction line. Users can select transaction type, date, product or cost group, customer or supplier, value, quantity, payment method, due date, and notes. The system calculates each row based on the selected transaction type.

For revenue rows, the value increases recorded revenue and may reduce inventory if a product is sold. For inventory purchase rows, the value increases stock and may increase cash paid. For other expense rows, the value increases business expenses and cash paid but does not affect inventory quantity.

Before saving, the system should check every row. Rows with missing required fields, invalid numbers, or incomplete information should be corrected. Valid rows can be saved and processed together.

After users click Save All Transactions, the system updates dashboard data, transaction history, inventory quantity, cash paid, profit, and cash flow based on all valid rows. The Save Draft option allows users to keep unfinished entries and complete them later. The Delete Selected Rows option removes unwanted rows before saving.

10. Products and Inventory Page

The Products and Inventory page manages product records, stock quantity, inventory value, categories, suppliers, and stock status. The system displays products with information such as product name, category, current stock, stock threshold, cost price, selling price, and margin. Users can search, sort, filter, and customize visible columns. Current stock is calculated using: $\text{Current Stock} = \text{Opening Stock} + \text{Stock In} - \text{Stock Out}$. Inventory value is calculated using: $\text{Inventory Value} = \text{Current Stock Quantity} \times \text{Cost Price}$

The system compares current stock with the warning threshold. If the stock quantity is lower than or equal to the threshold, the product is marked as low stock. If current stock equals zero, the product is shown as out of stock.

The inventory summary cards show information such as total products, total stock, inventory value, low-stock items, and out-of-stock items. These values update when users add products, declare initial inventory, record stock-in, or record sales.

The action-needed panel helps users identify products or business areas that require attention. For example, it may show low-stock products, unpaid customers, or unpaid costs.

11. Add Product Page

The Add Product page allows users to create a new product profile. Users enter product name, unit of measurement, expected selling price, stock alert threshold, and category. Users can choose an existing category or create a new one. Categories help group similar products and support later analysis.

The system gives users two options: create the product first and add stock later, or stock in immediately. If users choose to create the product only, the system saves product information without increasing inventory quantity. The product will appear in the product list, but stock remains zero or empty until inventory is added later. If users choose to stock in immediately, the system asks for stock quantity, cost price per unit, supplier, stock-in date, payment method, and notes. After saving, the system creates the product profile and increases inventory quantity at the same time. If the stock-in is treated as a paid purchase, the amount is included in cash paid. The system also updates inventory value based on quantity and cost price.

VI. User Flow

1. Landing Page

The user journey begins on the PIXIU FLOW landing page. This page introduces the website as a simple financial management tool for small businesses. Visitors can read the main headline, product description, feature highlights, target user groups, and general information about the platform.

From this page, visitors have several possible actions. If they already have an account, they can click Login to access the system. If they are new users, they can select Sign Up or Try for Free to start the registration process. Visitors who want to understand the product before registering can choose Watch Demo.

The landing page does not require users to enter financial or business data. Its main function is to explain the product value and guide visitors to the correct next step.

2. Login Page

Returning users enter the login page when they want to access their existing PIXIU FLOW account. The login form asks for a username, email address, or phone number, together with the password.

After filling in the required information, the user clicks Login. The system checks whether the account details are correct. If the information is valid, the user is taken to the dashboard. If the username, email, phone number, or password is incorrect, the login page remains visible and the user needs to correct the information.

The page also supports several additional actions. The user can tick Remember Login to make future access faster. The password can be shown or hidden using the eye icon. If the user forgets the password, the Forgot Password link allows them to start the recovery process. The user can also choose Continue with Google as an alternative login method.

If a visitor does not have an account yet, they can click Register Now to move to the sign-up flow.

3. Sign Up and Onboarding Flow

New users start the onboarding process by creating an admin account. In the first step, they enter their email address or phone number, the business owner's name, a password, and password confirmation. The confirmation field helps prevent password setup errors.

After completing the account form, the user continues to the business profile step. In this step, they enter the business or store name, select the business type, choose the main currency, and select the country. This information allows PIXIU FLOW to prepare the workspace for that specific business.

If the onboarding process includes category setup, the user can also define product categories and expense categories. Product categories will later be used for inventory and sales classification, while expense categories will be used when recording business costs.

After the required setup steps are completed, the system creates the user account and business profile. The user can then enter the dashboard or continue setting up business data.

4. Initial Inventory Declaration

After entering the system for the first time, the user may be guided to declare initial inventory. This step is designed for goods that the business already had before starting to use PIXIU FLOW. The user opens the Initial Inventory Declaration page and enters product information in a table. For each product, they fill in the product name, category, unit, selling price, estimated cost price, initial stock quantity, stock date, and low-stock alert threshold.

If the business has several existing products, the user can add more rows to the table. If a row is entered incorrectly, it can be deleted before saving. Once the information is complete, the user clicks Save Changes.

After saving, the system creates product records and opening stock balances. These products then appear in the inventory section. The important point is that this step does not create a new purchase transaction and does not create cash outflow. It only records stock that already existed before the user started using the website.

If the user has just bought new goods, they should choose Go to Stock-In instead. If they only want to create product profiles without stock quantity, they can choose Add Product Only. If they do not want to complete this step immediately, they can click Skip and return to it later.

5. Dashboard Flow

After login or onboarding, the user enters the dashboard. This page gives an overview of business performance. It shows key indicators such as recorded revenue, gross profit, net profit, cash paid, and net cash flow.

If the business has not entered any data yet, the dashboard displays empty values and a getting-started guide. The guide suggests three first actions: declare existing stock, record the first transaction, and view more indicators. This helps new users understand what to do next instead of seeing an empty dashboard without direction.

The user can filter dashboard data by selecting All, Today, 7 Days, This Month, This Year, or a custom date range. After choosing the period and clicking Apply, the system refreshes all dashboard figures and charts based on the selected time range.

The dashboard also includes trend charts and quick statistics. The user can review financial movements over time and check summary indicators such as average revenue, average cost, gross profit margin, and transaction count.

From the dashboard, the user can move to other main sections, including transaction recording, transaction history, product and inventory management, and additional analysis.

6. Recording a Revenue Transaction

To record sales, the user opens the transaction recording page and selects the New Revenue tab. The form allows the user to enter customer information, transaction date, product details, payment method, and notes. Inside the order detail section, the user selects or enters the product sold, the unit selling price, and the quantity. The system calculates the line total automatically. If the sale includes more than one product, the user can add another product line.

After checking the information, the user clicks Save Transaction. The system saves the revenue transaction, updates transaction history, recalculates dashboard figures, and reduces inventory quantity if the sold product is linked to stock. If the user decides not to save the transaction, they can click Cancel and leave the form without changing the data.

7. Recording an Inventory Purchase Cost

When the user wants to record a cost related to buying goods, they open the transaction recording page and select New Cost. Then they choose Inventory Purchase Cost. The form asks for the transaction date, supplier, purchased product, purchase unit price, quantity, payment method, payment due date if needed, and notes. The user enters the purchase details in the purchase table. If several products are bought in the same transaction, more product lines can be added.

After the form is completed, the user clicks Save Cost. The system calculates the purchase cost, increases inventory quantity, updates inventory value, records the transaction in history, and updates dashboard indicators. If the purchase has been paid, the amount is counted as cash paid. If payment will be made later, the due date is stored for tracking.

8. Recording Other Expenses

If the cost is not related to buying inventory, the user selects Other Expense instead of inventory purchase cost. The form changes to show expense details. The user enters the transaction date, expense group, amount, payment method, payment due date if needed, and notes. Examples of other expenses include electricity, rent, delivery fees, packaging, marketing, wages, and other operating costs. The user can add more expense lines if one transaction includes several cost items. The system adds all lines together and displays the total cost.

After clicking Save Cost, the system stores the expense in transaction history and updates dashboard results. Other expenses affect costs, cash paid, net profit, and cash flow, but they do not change inventory quantity. If the user does not want to keep the entry, they can click Cancel.

9. Multiple Transaction Entry

When several transactions need to be recorded at the same time, the user opens the multiple transaction entry page. This page uses a table format so that many transactions can be entered more quickly. Each row represents one transaction or transaction line. The user selects the transaction type, date, product or cost group, customer or supplier, value, quantity, payment method, due date, and notes. The user can add one row manually or add ten rows at once. If some rows are no longer needed, they can be selected and deleted. If the work is not finished, the user can save the table as a draft. When the user clicks Save All Transactions, the system checks the entered rows. Valid transactions are saved, while incomplete or incorrect rows need correction.

After saving, the system updates the dashboard, transaction history, inventory records, cash paid, profit, and cash flow based on the transaction types entered.

10. Transaction History Flow

The transaction history page allows users to review all recorded business activities. It includes revenue transactions, inventory purchase costs, and other expenses. The user can filter transactions by time period, such as all records, today, 7 days, this month, or a custom date range. They can also search by product, product code, partner, customer, supplier, value, or keyword. Transaction type tabs allow the user to view all transactions, revenue only, inventory purchases, or other expenses.

The table shows important transaction details such as type, date, content, partner, quantity, value, payment status, and action buttons. If a transaction contains incorrect information, the user can click the edit icon. After changes are saved, the system recalculates affected figures such as revenue, cost, cash paid, profit, cash flow, and inventory quantity. If a transaction was entered by mistake, the user can delete it. The system removes that transaction and updates the related dashboard and inventory figures. The transaction history page also includes summary cards. These cards change based on the selected filters, allowing users to review business performance for a specific period or transaction group.

11. Products and Inventory Flow

The Products and Inventory page is used to manage product records, stock quantity, inventory value, categories, suppliers, and stock status. The user can search for products, filter product information, sort table columns, and customize which columns are displayed. The product table shows information such as product name, category, current stock, warning threshold, cost price, selling price, and margin. If the user wants to add more inventory, they can click Stock-In. If they want to create a new product, they can click Add Product.

The inventory page also shows summary cards, including total products, total stock, inventory value, low-stock items, out-of-stock items, or products needing attention. These indicators help the business owner quickly understand the current inventory condition. If stock quantity is lower than or equal to the warning threshold, the system marks the product as low stock. If stock quantity is zero, the product is shown as out of stock. Based on this information, the user can decide whether to restock, review the product, or adjust purchase planning.

12. Add Product Flow

When the user wants to create a new product, they open the Add Product page. The form asks for product name, unit of measurement, expected selling price, stock alert threshold, and category. The user can choose an existing category or create a new one. After entering the product information, they choose whether to create only the product profile or stock in immediately. If the user selects Create Product Only, the system saves the product information without increasing inventory. The product appears in the product list, but stock quantity remains zero or empty until stock is added later. If the user selects Stock In Now, the form asks for stock quantity, cost price per unit, supplier, stock-in date, payment method, and notes.

After saving, the system creates the product and records the stock-in information at the same time. This flow helps users create products flexibly, depending on whether they already have stock information available.

13. Stock-In Flow

The stock-in flow is used when the business buys new goods or adds more inventory after starting to use PIXIU FLOW.

The user selects the product, enters the purchase quantity, unit cost, supplier, stock-in date, payment method, payment due date if necessary, and notes. After saving, the system increases stock quantity and updates inventory value. If the purchase is paid immediately, the amount is included in cash paid. If payment is due later, the due date helps the user track future payment obligations. Stock-in is different from initial inventory declaration. Initial inventory records goods that already existed before using the software and does not affect cash paid. Stock-in records new purchases and can affect both inventory and cash flow.

14. Report and Business Review Flow

After transactions and inventory records are entered, the user can return to the dashboard or report sections to review business performance.

The user selects a time period and checks indicators such as revenue, cost, profit, cash paid, net cash flow, inventory value, and transaction count. Charts and summary cards help the user understand whether the business is improving or facing problems.

Based on the results, the business owner can make decisions such as restocking products, reducing unnecessary expenses, adjusting selling prices, reviewing suppliers, or focusing on more profitable products.

VII. Output

1. Landing Page	
Element / Feature	Description
Navigation Bar	Deep crimson (#8B1A1A) full-width top bar with PixiuFlow logo (gold Pixiu mascot + wordmark) on the left and two outlined buttons: 'Đăng nhập' (Login) and 'Đăng ký' (Sign Up) on the right.
Hero Badge	Small badge 'Phần mềm quản lý tài chính phổ biến nhất' (Most popular financial management software) with a star icon, positioned above the main headline to emphasize the badge.
Hero Headline	Large bold Vietnamese headline: 'Quản lý tài chính đơn giản cho doanh nghiệp nhỏ' displayed in deep crimson at approximately 56px, left-aligned.
Hero Subtext	Supporting paragraph in body size explaining the core value proposition: track income/expenses, manage inventory, and view profit reports in minutes without deep accounting knowledge
Primary CTA Button	Filled crimson button labeled 'Dùng thử miễn phí' (Try for free): primary action, positioned below the subtext
Secondary CTA Button	Outlined button with play icon labeled 'Watch demo': secondary action, placed beside the primary CTA
Feature Microcopy	Four short trust-building feature callouts below the CTAs: automatic expense recording, real-time inventory tracking, instant profit/loss reporting, and Vietnamese language support
Hero Visual	Right-side mockup of the PixiuFlow dashboard displayed inside a laptop frame showing bar charts in crimson and cream, providing a product preview without requiring users to sign up
Target Audience Label	Full-width tagline: 'Được thiết kế dành riêng cho hộ kinh doanh và doanh nghiệp siêu nhỏ tại Việt Nam'; emphasizing the product's Vietnamese small business focus
Business Type Row	Four illustrated business type cards: Cửa hàng bán lẻ (Retail), Quán cà phê, nhà hàng (Café/Restaurant), Shop online, Dịch vụ cá nhân (Personal Services); each with a custom illustration

Feature Highlights Section	Five core feature cards with icons: auto expense/income recording, inventory tracking, profit/loss reporting, smart alerts, and (implied) data security
About Us Section	Three-column layout: 'PixiuFlow là gì?' (product description), 'Chúng tôi là ai?' (team background ‘University students from FTU’), 'Sứ mệnh' (mission statement with brand tagline)
Trial Banner	Full-width crimson-background CTA banner with '14 NGÀY Dùng thử miễn phí' (14-day free trial) button; placed before the footer as a final conversion prompt
Brand Tagline	Closing statement: 'Pixiu Flow - Từ số liệu đến hành động' (From data to action); displayed as a testimonial-style quote in the About section

2. Login Page	
Element / Feature	Description
Overall Layout	Full-screen split layout: deep crimson left panel (40% width) with brand content and Pixiu mascot illustration, warm cream right panel (60% width) with the login form
Left Panel - Brand Content	PixiuFlow logo at top, bold white headline 'Quản lý doanh thu & lợi nhuận doanh nghiệp dễ dàng hơn', supporting subtext about the platform's value, and two trust badges: 'Bảo mật dữ liệu' and 'Phân tích chính xác'
Left Panel - Mascot	Full illustrated Pixiu mascot (gold and red, Vietnamese mythological lion-dog character) positioned at the bottom-center of the left panel
Left Panel - Footer	Small copyright text at the very bottom: '© 2026 PIXIU FLOW - Nền tảng quản lý tài chính doanh nghiệp thông minh. Phiên bản 1.0-beta'
Form Container	Rounded card with cream (#FDF6EC) background, slight drop shadow, centered on the right panel

Form Title	'Đăng nhập' heading with subtitle 'Chào mừng bạn quay trở lại. Vui lòng nhập thông tin.'
Username Field	Input labeled 'Tên đăng nhập / Email / Số điện thoại' with envelope icon and placeholder text 'username hoặc email@company.com'
Password Field	Input labeled 'Mật khẩu' with lock icon, masked entry (dots), eye icon toggle for show/hide, and 'Quên mật khẩu?' link in crimson on the right
Remember Me	Checkbox with label 'Ghi nhớ đăng nhập' (Remember login)
Primary Login Button	Full-width filled crimson button with white text 'Đăng nhập'
Divider	Thin horizontal line with centered text 'Hoặc Đăng nhập với' (Or Sign in with)
Google SSO Button	Full-width outlined button with Google icon and label 'Tiếp tục với Google'
Sign Up Redirect	Bottom text: 'Chưa có tài khoản? Đăng ký ngay >' with 'Đăng ký ngay' in bold and underlined
Footer Links	Three small links at the very bottom: Điều khoản dịch vụ / Chính sách bảo mật / Hỗ trợ khách hàng

3. Sign Up / Onboarding Flow

Element / Feature	Description
Overall Layout	Full-screen layout with crimson left sidebar showing step progress, and cream main content area on the right spanning the full height
Left Sidebar - Steps	Vertical step progress indicator with 4 steps: 1. Tạo tài khoản (Create account) 2. Hồ sơ doanh nghiệp (Business profile) 3. Danh mục hàng bán (Product categories) 4. Danh mục chi phí (Expense categories). Active step highlighted with a circle; completed steps show a green checkmark

Page Header	Top breadcrumb: 'Thiết lập hệ thống > Trình hướng dẫn PIXIU FLOW' with language selector (Tiếng Việt VN) and support phone number (1900 8888) on the right
Welcome Headline	Large heading 'Chào mừng đến với PIXIU FLOW' with subtitle 'Thiết lập doanh nghiệp của bạn chỉ trong vài phút để bắt đầu quản lý tài chính thông minh.'
Step 1 - Account Creation	Card section labeled 'Bước 1: Tạo tài khoản quản trị': two-column form with Email/Phone field and Create Password field, plus Confirm Password field below; Google quick-signup button and 'Tiếp tục thiết lập' (Continue setup) button in crimson
Step 2 - Business Profile	Card section labeled 'Bước 2: Hồ sơ doanh nghiệp', fields for Business Name/Store, Business Type (dropdown: Bán lẻ/Retail), Currency Unit (VND - Việt Nam Đồng), Country (Việt Nam); 'Lưu và tiếp tục' button
Pixiu Mascot - Decorative	Small Pixiu mascot illustrations placed at the left edge of each step card, peeking into the form, add personality and brand consistency
Completion Screen	Final section with a circular green checkmark icon, heading 'Tuyệt vời! Bạn đã sẵn sàng bắt đầu', confirmation text, two buttons: 'Bỏ qua danh mục và tiếp tục' (outlined) and 'Vào dashboard ngay' (filled crimson)
Terms Notice	Small text below completion CTA: 'Bằng cách nhấn "Vào dashboard", bạn đồng ý với Điều khoản dịch vụ của PIXIU FLOW.'
Bottom Bar	Fixed footer: '2026 Pixiu Flow - Nền tảng quản lý tài chính doanh nghiệp thông minh. Phiên bản 1.0 - BETA'
Logged-in User Indicator	Bottom-left of sidebar shows user avatar, name 'user', and status 'Đang thiết lập...'; confirming the account was successfully created in the previous step

4. Initial Inventory Declaration (Onboarding Step)

Element / Feature	Description
Overall Layout	Same crimson sidebar + cream content area as the onboarding flow, now showing step 'Hồ sơ doanh nghiệp' active in the progress indicator
Section Label	Small uppercase label 'Thiết lập ban đầu' above the heading
Page Heading	'Khai báo hàng ban đầu' with subtext explaining that this step is for declaring existing stock before starting to use PixiuFlow - It is not a new purchase transaction
Info Banner	Yellow-background notice: 'Hàng ban đầu chỉ dùng cho số hàng đã có trước khi dùng Pixiu Flow.' with explanation text and two action buttons: 'Đi tới Nhập kho' (outlined) and 'Chỉ thêm sản phẩm' (filled crimson)
Product Table - Headers	Seven columns: - Tên Sản Phẩm (Product Name, Required), - Danh Mục (Category With Dropdown), - Đơn Vị (Unit With Dropdown), - Giá Bán (Selling Price), - Giá Vốn Ước Tính Vnd (Estimated Cost Price), - Tồn Kho Ban Đầu (Initial Stock), - Ngưỡng Cảnh Báo (Alert Threshold)
Product Table - Rows	Editable inline rows with placeholder values (e.g: Váy, giày...); each row has a red delete button on the right; at least two rows shown by default
Add Row Button	Outlined pill button '+ Thêm dòng' (Add row) below the table allows users to add more products
Action Buttons	Two buttons bottom-right: 'Lưu tồn kho khởi đầu' (filled crimson - Save initial stock) and 'Bỏ qua' (Skip - outlined)
Pixiu Mascot	Pixiu mascot illustration positioned top-right of the content area, maintaining visual consistency with the rest of the onboarding flow

5. Dashboard / Bảng theo dõi	
Element / Feature	Description
Overall Layout	Two-panel layout: deep crimson left sidebar (navigation) + cream main content area; top bar with search, notification bell and user profile badge
Left Sidebar	PixiuFlow logo at top, navigation items: • Trang chủ, • Bảng theo dõi (active, highlighted with cream background), • Giao dịch (with sub-items: Ghi nhận một giao dịch, Ghi nhận nhiều giao dịch, Xem lịch sử giao dịch), • Hàng tồn kho (Inventory storage) • Phân tích thêm (Additional analysis)
Top Bar	Full-width cream bar with search field 'Tìm kiếm giao dịch, đối tác...' on the left, notification bell icon, and user profile (avatar + name 'user' + role 'Chủ doanh nghiệp') on the right
Page Title	'Bảng theo dõi' as main heading with subtitle 'Theo dõi doanh thu, chi phí và hiệu quả kinh doanh theo thời gian thực'
First-time Onboarding Checklist	Cream card labeled 'Bắt đầu với Pixiu Flow' with 'Ân hướng dẫn này' link. Three numbered steps with arrows: Khai báo hàng ban đầu → Ghi nhận giao dịch đầu tiên → Xem báo cáo. Step 3 is highlighted as active (green background). Each step has a title, short description, and clickable link
Date Filter Bar	Label 'Bộ Lọc Dữ Liệu Cho Toàn Bộ Bảng' with five period buttons: Tất cả / Hôm nay / 7 ngày / Tháng này (active, crimson) / Năm nay, plus a custom date range picker (01/06/2026 - 06/06/2026) and 'Áp dụng' (Apply) button
KPI Card Row 1	Three cream metric cards: • Doanh thu ghi nhận (Recorded Revenue) with trend-up icon in green, • Lợi nhuận gộp (Gross Profit) with calendar icon, • Lợi nhuận thuần (Net Profit) with clock icon. All showing with empty-state description below each value

KPI Card Row 2	Three more cream metric cards: • Tiền đã thu (Cash Collected) with dollar icon in teal, • Tiền đã chi (Cash Spent) with wallet icon in orange, • Dòng tiền thuần (Net Cash Flow) with briefcase icon in green.
Empty State - Chart Area	Large cream card below the KPI cards with a bar chart icon and text 'Phân Tích Chi Tiết Sẽ Hiện Thị Ở Đây => Khi bạn có dữ liệu giao dịch, các biểu đồ xu hướng sẽ hiện ra' with a loading progress bar; guiding new users to start recording transactions
Footer	Full-width footer: '2026 Pixiu Flow - Nền Tảng Quản Lý Tài Chính Doanh Nghiệp Thông Minh. Phiên bản 1.0 - BETA' with Hỗ trợ, Điều khoản, Bảo mật links on the right

6. Transaction Record - Revenue (Ghi nhận giao dịch - Doanh thu)	
Element / Feature	Description
Page Title	'Ghi nhận giao dịch' as main heading with subtitle 'Ghi nhận, theo dõi và quản lý giao dịch doanh thu & chi phí một cách chuyên nghiệp.'
Tab Toggle	Two tabs at the top of the form: 'Doanh thu mới' (filled crimson, active) and 'Chi phí mới' (outlined), switching between revenue and expense entry modes
Transaction ID	Auto-generated transaction ID displayed on the right: 'Mã GD: #TXN-1203'
Customer Field	Dropdown labeled 'Khách Hàng (Tùy Chọn)' with placeholder 'Chọn khách hàng hoặc đối tác' and a search/select interaction
Transaction Date	Date picker labeled 'Ngày Giao Dịch (DD/MM/YYYY)' pre-filled with '30/11/2024' and a calendar icon
3-Level Category	Three sequential dropdowns: • DANH MỤC CẤP 1 (e.g: 'Đồ ăn'), • DANH MỤC CẤP 2 (e.g: 'Mì tôm'), • DANH MỤC CẤP 3 (e.g: 'Mì koreno') This allows precise categorization of each transaction

Order Detail Table	Dashed-border card labeled 'Chi tiết đơn hàng' with a note 'Đang nhập 01 mục'. Table columns: - Tên Sản Phẩm (Product), - Đơn Giá Vnd (Unit Price), - Số Lượng (Quantity), - Thanh Toán Vnd (Total). Default first row with 'Mì tôm' as product, unit price 0, quantity 1, total 0. Delete icon on the right
Add Product Row	‘Thêm sản phẩm khác cho giao dịch này’ below the table for adding multiple line items to one transaction
Payment Method	Dropdown labeled 'Phương Thức Thanh Toán' with default value 'Chuyển khoản Ngân hàng' (Bank Transfer)
Invoice Upload	Dashed upload area labeled 'Ảnh Hóa Đơn (Tùy Chọn)' with text 'Kéo thả hoặc Click để tải lên'; optional invoice image attachment
Total Display	Bottom of form: 'Tổng Cộng Doanh Thu:' label with large crimson value '25.000.000 VND'
Form Action Buttons	Two buttons: 'Hủy' (Cancel, outlined) and 'Lưu giao dịch' (Save Transaction, filled crimson)
KPI Sidebar - Revenue	Right panel card: 'Tổng doanh thu hôm nay' = 45,200,000 VND with green trend-up arrow and '+12.5% so với hôm qua'
KPI Sidebar - Expense	Right panel card: 'Tổng chi phí hôm nay' = 12,850,000 VND with red trend-down arrow and '+4.2% so với hôm qua' (shown as increase in expenses)
KPI Sidebar - Profit	Right panel card: 'Lợi nhuận hôm nay' = 32,350,000 VND with orange wallet icon and '+8.1% so với hôm qua'
Recent Transaction Feed	Bottom of sidebar: 'Giao Dịch Gần Đây' section with two recent entries: 'Thanh toán Hosting AW...' +12M (29/11/2024) and 'Thanh toán...' -3.5M (28/11/2024), color-coded green/red by type

7. Transaction Record - Expense (Ghi nhận giao dịch - Chi phí)

Element / Feature	Description
Tab Toggle	'Chi phí mới' tab is now active (filled crimson); 'Doanh thu mới' becomes outlined
Expense Sub-type Toggle	Two radio-style buttons below the tab: 'Chi phí nhập hàng' (Inventory purchase expense) and 'Chi phí khác' (Other expense, active by default); determines the form structure shown below
Transaction Date	Date field (same as revenue screen) for recording when the expense occurred
Expense Detail Table	Table with three columns: # (row number), Nhóm Chi Phí (Expense Group- dropdown for category selection), Số Tiền (VND) (Amount), Ghi Chú (Note- free text). One default empty row; row can be deleted
Add Expense Row	'Thêm chi phí khác' link for adding multiple expense line items
Total Display	'Tổng Cộng Chi Phí:' with large crimson value '0 VND' (empty state)
Form Action Buttons	'Hủy' (Cancel) and 'Lưu chi phí' (Save Expense, filled crimson)
KPI Sidebar	Same three KPI cards as the revenue screen: - Profit today (32,350,000 VND), - Total expenses today (12,850,000 VND), - Total revenue today (45,200,000 VND) This provides real-time business context while entering expenses
Recent Transaction Feed	Same feed as revenue screen, which is consistent across both transaction entry modes

8. Bulk Transaction Entry (Ghi nhận nhiều giao dịch)

Element / Feature	Description
Page Title	'Ghi nhận nhiều giao dịch' with subtitle 'Nhập nhiều giao dịch doanh thu hoặc chi phí cùng lúc để tiết kiệm thời gian.'
Action Bar	Row of four buttons: 'Thêm dòng' (Add row, outlined), 'Import Excel' (Import from Excel), 'Tải file mẫu' (Download template), 'Xóa dòng' (Delete row, red text)
Bulk Table - Columns	Eight columns: - Loại Giao Dịch (Transaction Type - Doanh Thu/Chi Phí Dropdown), - Danh Mục Cấp 1 (Category L1 Dropdown), - Danh Mục Cấp 2 (Category L2 With Inline Suggestion), - Giá Sản Phẩm (Vnd) (Unit Price), - Số Lượng (Quantity), - Khách Hàng (Customer/Partner), - Thành Tiền (Total - Auto-Calculated)
Table - Row Examples	- Row 1: Doanh thu / Bán hàng & Dịch vụ / Gói = 15,000,000 / 1 / Công ty Alpha Tec = 15,000,000đ. - Row 3: Chi phí / Vận hành / Tiền điện + Văn phòng Q1 / 2,500,000 / 1 / EVN HCM = 2,500,000đ
Inline Validation	Row 4 shows a negative value '-3500000' highlighted in a red input field with an error indicator - real-time validation prevents invalid data entry
Inline Suggestion	Row 2 shows a teal dropdown suggestion pill appearing below a category field, offering auto-complete options as the user types
Running Total Footer	Fixed bottom bar: 'Tổng Cộng Dòng Hợp Lệ' = 24,000,000đ with 'Hủy' (Cancel) and 'Lưu nhập' (Save entries) buttons, plus 'Lưu trữ và giao dịch' button on the far right
Pixiu Mascot	Pixiu mascot illustration sits at the bottom-right corner of the table area, maintaining brand personality

9. Transaction History (Lịch sử giao dịch)	
Element / Feature	Description
Page Title	'Lịch sử giao dịch' with subtitle 'Xem lại toàn bộ doanh thu, chi phí và nhập hàng. Có thể sửa hoặc cập nhật dữ liệu ngay.'
Quick Add Buttons	Two crimson buttons in the top-right: 'Thêm giao dịch mới' and 'Thêm giao dịch mới' (revenue vs expense variants) This allows fast transaction creation from the history view
Search & Filter Bar	Text search field 'Tìm theo tên sản phẩm, nội dung giao dịch, khách hàng...' on the left; date range picker 'Từ ngày → Đến ngày' on the right
Summary Cards	Three KPI cards: - Tổng tiền vào (Total In) = 395,000đ with green trend +22.5%, - Tổng tiền ra (Total Out) = 261,986đ with yellow trend +4.2%, - Chênh lệch (Difference) = 133,014đ with a blue exchange icon labeled 'Lợi nhuận dương' (Positive profit)
Filter Tabs	Five tab filters: - Doanh thu (Revenue) - Chi phí (Expense) - Nhập hàng (Import) - Tìm nhanh giao dịch... (Quick search) - Ngày gần đây (Recent dates)
Transaction Table - Headers	Seven columns: - Loại (Type), - Ngày (Date), - Nội Dung (Description), - Đối Tác (Partner), - Số Lượng (Quantity), - Tổng Tiền (Vnd) (Amount), - Thao Tác (Actions)
Transaction Table - Type Badges	Color-coded transaction type badges: green pill 'Doanh thu' for revenue entries, orange/yellow pill 'Nhập hàng' for inventory purchases, making transaction types instantly scannable

Transaction Table - Rows	Example rows: • 26/04/2026 Mì tôm / Linh / 50 units / +300,000đ (Doanh thu); • 26/04/2026 Mì tôm / B / 50 units / -200,000đ (Nhập hàng); • 19/04/2026 Mì tôm / Khách lẻ / 5 units / +35,000đ. Revenue amounts shown in green, expense/import amounts in red
Amount Color Coding	Positive amounts (revenue) displayed in green; negative amounts (expense/import) displayed in red => Instant visual distinction

10. Inventory Management (Hàng tồn kho)

Element / Feature	Description
KPI Summary Row	Four cream summary cards at the top: • Tổng sản phẩm kinh doanh (Total products) = 1,284 with +42% growth badge; • Tổng tồn kho (Total stock units) = 42,500; • Giá trị tồn kho (Inventory value) = 5.24B VND with +%; • Sắp hết hàng (Low stock) = 18 items with red badge
Section Title	'Danh sách sản phẩm' (Product List) with '+ Thêm sản phẩm' (Add Product) crimson button on the right
Search & Filter Bar	Search field 'Tìm kiếm sản phẩm...' with 'Danh mục' (Category filter) and 'Trạng thái' (Status filter) buttons
Sort Options	Two sort toggles: 'Sắp xếp số lượng' (Sort by quantity) and 'Sắp xếp giá trị' (Sort by value)
Product Table - Headers	Eight columns: • Tên sản phẩm (Product name), • Mã SP (SKU), • Danh mục (Category), • Số lượng (Quantity), • Giá vốn (Cost price), • Giá bán (Selling price), • Trạng thái (Status badge), • Hành động (Actions: edit/delete icons)

Product Table - Sample Rows	iPhone 15 Pro Max 256GB (SKU: APP-256) / Điện thoại / 124 units / 26.5M / 32.0M / Còn hàng (green badge). MacBook Air M2 13 inch / Laptop / 12 units / 24.2M / 28.9M / Hàng mới (blue badge). AirPods Pro Gen 2 / Phụ kiện / 0 units / 4.1M / 5.6M / Hết hàng (red badge).
Status Badges	Four distinct status badges: 'Còn hàng' (green - In stock), 'Hàng mới' (blue - New arrival), 'Hết hàng' (red - Out of stock), 'Sắp hết' (amber - Low stock)
Category Panel	'Phân loại' section with 'Thêm' button - table showing category name, product count, and action menu.
Supplier Panel	'Nhà cung cấp' section with 'Thêm' button - table showing supplier name, phone number, and action menu: Apple Vietnam (028 1234xxx), FPT Trading (024 0678xxx), Digiworld (028 9101xxx)
Stock Alert Section	'Cảnh báo hàng tồn' with four color-coded alert cards: - Hết hàng 08 SP (red), - Sắp hết 12 SP (orange), - Tồn kho lâu 24 SP (blue), - Cần ý nhập 05 SP (green). Alert table below: product name, current quantity, threshold, alert type, suggested action
Lot Tracking Table	'Lô hàng & Theo dõi' with 'Tạo lô hàng mới' button. Table: Mã lô (Lot ID), Sản phẩm, Nhà cung cấp, Ngày nhập, Hạn dùng, Trạng thái. Sample lots with status badges: Còn hạn (In date, green), Sắp hết hạn (Near expiry, amber), Hết hạn (Expired, red)
Right Sidebar - Best Sellers	'Sản phẩm bán chạy nhất': iPhone 15 Pro Max 64 đơn, AirPods Pro Gen 2 156 đơn, Cáp sạc USB-C 60W 362 đơn, MacBook Air M2 22 đơn, Ốp lưng Silicone 215 đơn
Right Sidebar - High Margin	'Danh mục biên lợi nhuận cao': Phụ kiện Apple 42%, Dịch vụ bảo mật 35%, Thiết bị âm thanh 28%
Right Sidebar - Inventory Value	Total inventory value card: 5,248,000,000 VND (approximately 12 billion at selling price) with +13% trend
Right Sidebar - Top Suppliers	'Nhà cung cấp lớn nhất': Apple Vietnam 3.2B, FPT Trading 1.1B, Digiworld 450M

11. Advanced Analytics / Phân tích chuyên sâu	
Element / Feature	Description
Page Title & Controls	'Phân tích chuyên sâu' heading with period filter dropdown ('Tháng này'), category filter ('Danh mục'), and 'Xuất dữ liệu' (Export data) crimson button in the top bar
Revenue Trend Chart	Area chart with crimson fill labeled 'Phân tích doanh thu' - X-axis shows months T6-T11, Y-axis 0-320,000,000 VND. Green bar chart overlaid showing profit per period. Toggle dropdown: 'Theo danh mục' with VND/% switch
Revenue Breakdown Table	Right side of the revenue section: Category table with columns Danh mục / Doanh thu / Tăng trưởng. Rows: Đồ công nghệ 145M +12%, Thời trang Nam 89M +5.4%, Phụ kiện 52M -2.1% (shown in red). Total row: 286,000,000 VND +8.2%
Expense Donut Chart	'Phân tích chi phí' section with a multi-color donut chart showing total of 226.5M VND. Center label shows 'Tổng chi phí 226.5M'. Toggle: Theo danh mục, VND/%
Expense Breakdown Table	Right of the donut: Danh mục / Giá trị VND / % / Xu hướng. Rows: Giá vốn hàng bán 125,000,000 (55%), Chi phí nhân sự 45,000,000 (20%), Mặt bằng & Điện nước 25,000,000 (11%), Marketing & Quảng cáo 18,000,000 (8%), Chi phí khác 13,500,000 (6%)
Profit Analysis Table	'Phân tích lợi nhuận' card with 'Tỉ suất: 24.5%' badge. Table: Danh mục / Lợi nhuận / Biên LN. Rows: Điện tử 32.4M / 22%, Sức khỏe 18.2M / 35%, Gia dụng 12.5M / 18%
Inventory Health Panel	'Tổng quan hàng tồn kho' card with 'Sắp hết: 12 mặt hàng' alert badge. Two summary stats: TỒN KHO 2,450 units / NGÀY TỒN TB 18 days. Stock status table: Thời trang Nam 12 ngày ỔN ĐỊNH (green), Thời trang Nữ 28 ngày CẢNH BÁO (orange), Phụ kiện túi xách 45 ngày TỒN LÂU (red)
Business Rankings Sidebar	Right panel titled 'Thông Tin Xếp Hạng' with 6 ranked insights:
- Best-selling Product	iPhone 15 Pro Max 256GB - 42 đơn (42 orders)

- Highest Margin Category	Thực phẩm chức năng Nhật - 38.5% gross margin
- Slowest-moving Item	Ốp lưng Leather Case (Vàng) - 62 ngày (62 days held)
- Current Cash Balance	1,250,000,000 VND - labeled 'Ổn định' (Stable)
- Monthly Net Cash Flow	+185,400,000 VND - 12% vs previous month
- Largest Supplier	Công ty TNHH MTV Alpha - 1.2B VND in total orders
Export Controls	Period filter and category filter are applied globally across all panels. 'Xuất dữ liệu' button allows full report export to PDF or Excel format for offline business review

12. Interactive Prototype	
Element / Feature	Description
Tool	Figma - all screens linked with interactive connections, transitions, and overlay behaviors
Journey 1 - New User Onboarding	Landing Page (CTA click) → Login Page → Sign Up (account creation) → Business Profile Setup → Initial Inventory Declaration → Dashboard (first-time empty state with onboarding checklist)
Journey 2 - Daily Transaction Recording	Dashboard (click 'Ghi nhận một giao dịch') → Transaction Record Revenue tab (fill form) → Save → Transaction History (new entry appears) → Dashboard (KPI cards update)
Journey 3 - Expense Recording	Transaction Record (click 'Chi phí mới' tab) → Expense type toggle (Chi phí nhập hàng vs. Chi phí khác) → Fill expense detail table → Save → Transaction History
Journey 4 - Financial Analysis	Dashboard (Bảng theo dõi) → date filter interaction → trend chart populates → click 'Phân tích thêm' in sidebar → Advanced Analytics screen → category filter → export button
Transitions	Page transitions use smooth slide or fade animations consistent with a web application feel; sidebar navigation highlights update on each screen change

Overlay Behaviors	Dropdown menus, date pickers, and modal dialogs behave as overlays; clicking outside closes them; consistent with standard web UI interaction patterns
Empty States	Dashboard and Transaction History screens include designed empty states with instructional text and CTA links; guiding first-time users to take their first action
Visual Consistency	All prototype screens use the crimson-and-cream design system, the same Pixiu mascot placement, identical sidebar navigation, and matching KPI card layouts - the product looks and feels unified across all journeys
Demo Usage	The prototype was used in the final project demo session to walk evaluators through the complete user experience without requiring a working codebase

VIII. Important Design Choices

1. Why did your team choose this problem?

Our team observed that most small businesses in Vietnam still record income and expenses using notebooks or Excel spreadsheets. As a result, many business owners do not know whether they are actually making a profit or a loss in real time. Existing accounting software, such as MISA or Fast Accounting, is designed for companies with professional accountants and is often too complex for small retailers or café owners. Therefore, we decided to address this gap by creating a tool that is simple enough for business owners to use independently while still providing accurate profit and loss insights.

2. Why did you choose this target user group?

We identified small and micro-business owners in Vietnam as our primary target users, specifically the four business types highlighted on our Landing Page: Retail stores, cafés/restaurants, online shops, and personal service providers.

These users have genuine financial management needs but typically lack the resources to hire accountants. They are also unfamiliar with complex financial terminology and usually rely on basic computers or smartphones. For this reason, we designed the interface entirely in plain Vietnamese, avoided technical accounting knowledge, and created a workflow that can be completed in just a few simple steps.

3. Why did you choose a web app interface with a fixed sidebar?

We chose a web application layout with a fixed crimson sidebar rather than a mobile-first design or a simple landing page for two main reasons.

First, our target users often work at cashier counters or desks when recording transactions, making a desktop interface more suitable for entering large amounts of information efficiently.

Second, a dashboard structure with a persistent sidebar allows users to switch quickly between features such as Transactions, Inventory, and Analytics without losing context. This is particularly important when users need to review historical records while simultaneously entering new transactions.

4. Why did you choose crimson red as the primary color?

This was a deliberate cultural design decision. In Vietnamese and Chinese cultures, red symbolizes prosperity, luck, and business success, making it highly relevant for a financial management application. Our mascot, Pixiu, is a mythical creature believed to attract and preserve wealth, directly reinforcing the product's core value proposition. Combining a crimson sidebar with a cream-colored content area creates a visual identity that is both professional and welcoming, distinguishing PixiuFlow from traditional accounting software that often relies on cold blue color schemes.

5. Why did you implement a three-level transaction categorization system?

As shown in the Transaction Record screen, we designed a three-level category hierarchy (Level 1 → Level 2 → Level 3, for example: Food → Instant Noodles → Koreno Noodles). This decision was driven by real business needs. Small business owners often sell a wide variety of products and need to identify which products generate the most profit. A single-level category structure would be too broad for meaningful analysis. The three-level hierarchy enables detailed drill-down analysis at the stock-keeping unit level within the Advanced Analytics module, providing insights that simple tools like Google Sheets cannot easily deliver.

6. Why did you design a multi-step onboarding process instead of letting users explore the system themselves?

The Sign-Up process uses a guided onboarding flow with four clearly defined steps displayed in the sidebar:

- Account Creation,
- Business Profile Setup,
- Product Categories,
- Expense Categories.
- Initial Inventory Disclosure (Before Signing up)

We chose this approach because our target users are often unfamiliar with software products. If they entered an empty dashboard immediately, many would not know where to begin. The initial inventory setup is particularly important because skipping this step would result in inaccurate inventory tracking from the start. Even after registration, the dashboard's empty state includes a three-step checklist to continue guiding new users toward successful adoption.

7. Why did you add a Bulk Entry feature?

Our research revealed that many small business owners do not record transactions in real time. Instead, they often enter records at the end of the day or even at the end of the week. If the system only supported single-entry forms, entering 20-30 transactions at once would be time-consuming and discourage continued use.

The Bulk Entry feature, which includes a spreadsheet-style interface and Excel import functionality, aligns with users' existing workflows. Since many of them are already familiar with Excel, this feature allows them to migrate their data into PixiuFlow efficiently without re-entering everything manually.

8. Why did you exclude tax reporting and VAT invoice features?

Although tax reporting and VAT invoicing are important for larger businesses, we deliberately excluded these features because our primary users are micro-business owners, many of whom operate under household business models and do not issue VAT invoices regularly. Including such features would increase interface complexity and potentially confuse users. Instead, we provide an optional "Receipt Image" field within the transaction form, allowing users to attach receipts when necessary without requiring them to understand complex tax procedures.

9. Why did you display real-time KPIs directly on the transaction entry screen?

Rather than requiring users to navigate back to the Dashboard to view performance metrics, we placed key indicators such as Today's Revenue, Today's Expenses, and Today's Profit directly beside the transaction entry form. This decision was based on the insight that business owners want immediate feedback on their financial performance after recording a sale. By making these metrics visible in real time, we create a positive feedback loop: the more users record transactions, the more valuable insights they receive, which encourages continued engagement with the product.

10. Why did you separate Advanced Analytics from the main Dashboard?

The Dashboard is intentionally designed to provide a high-level overview, featuring six KPI cards and a simple trend chart. More detailed analyses, including category breakdowns, expense analysis, product rankings, and inventory alerts, are located within a dedicated Advanced Analytics module. We made this distinction to avoid overwhelming users when they access the system daily. The Dashboard should answer the question, "How is my business performing today?" within a few seconds. Questions such as "Why is this happening?" and "What actions should I take next?" belong in the Analytics section, where users can conduct deeper business reviews when they have more time available.

IX. Strengths of the Project

One of the biggest strengths of our project is that we successfully developed a solution that addresses a real-world problem: personal financial management. Instead of focusing solely on recording income and expenses, PixiuFlow was designed as a financial assistant that helps users better understand their spending habits, manage budgets, and make more informed financial decisions.

Another strength lies in the overall design consistency of the product. Throughout the development process, our team maintained a unified visual identity by using a consistent color palette, typography, and interface components. The red and gold theme was carefully selected to reflect the concept of prosperity and wealth while also aligning with the Pixiu mascot that inspired the product's branding.

The team also managed to create a complete user journey, including account registration, login, transaction management, budget tracking, financial reporting, and personalized financial suggestions. These features were connected through a logical workflow, allowing users to navigate the system intuitively and efficiently.

Furthermore, collaboration within the team was effective. Responsibilities were divided among members according to their strengths, while regular discussions ensured that all components of the project remained aligned with the overall vision. This helped us maintain a steady development pace and achieve our project goals within the given timeframe.

X. Current Limitations

Although the project achieved its primary objectives, there are still several limitations that could be addressed in future development.

First, PixiuFlow currently exists as a prototype rather than a fully functional application. While the user interface and workflows have been designed and demonstrated, the system has not yet been connected to a real database or external financial services. As a result, users cannot automatically synchronize transactions or access real-time financial data.

Second, the project mainly focuses on essential financial management functions. More advanced features such as AI-powered spending predictions, personalized investment recommendations, automated savings plans, and intelligent financial forecasting have not yet been implemented due to time and resource constraints.

From a design perspective, some screens remain relatively simple and could be further optimized to improve usability. Additional user testing may reveal opportunities to simplify navigation, reduce unnecessary steps, and improve the overall user experience.

Finally, because the project was developed within an academic setting and limited timeframe, the team was unable to conduct extensive testing with a large and diverse group of users. Therefore, further validation is needed to evaluate the effectiveness and practicality of the product in real-world scenarios.

XI. Lessons Learned and Recommendations for Future Students

The project also taught us how to transform a broad idea into a structured product. Starting from the general topic of personal finance management, we had to break the problem down into specific user needs such as expense tracking, budgeting, financial reporting, and spending analysis. This process helped us develop stronger problem-solving and product-thinking skills.

In addition, we gained experience in teamwork and project coordination. Working on a collaborative project required effective communication, task delegation, progress monitoring, and conflict resolution. We learned that maintaining regular communication is essential to ensuring that all team members move toward the same objective.

From a design perspective, we became more familiar with tools such as Figma and learned how to create user flows, wireframes, prototypes, and interface systems. These skills will be valuable for future projects involving digital product design and development.

Finally, preparing and presenting the project helped us improve our communication and presentation abilities. We learned how to explain our design decisions, justify product features, and demonstrate the value of our solution to different audiences.

Recommendations for Future Students

For future students who wish to build upon this project or develop similar products, our team would like to share several recommendations.

First, invest sufficient time in user research before starting the design phase. A strong understanding of user behavior and needs will provide a clearer direction for product development and help avoid unnecessary redesigns later.

Second, create detailed user flows and site maps early in the project. This will make it easier to organize features, maintain consistency across screens, and reduce confusion during implementation.

Third, if technical resources are available, consider moving beyond a prototype and developing a functional system with real data integration. Connecting the application to databases or financial APIs would significantly increase its practical value and realism.

Lastly, we strongly recommend starting the project as early as possible, maintaining regular progress reviews, and conducting user testing throughout the development process. Early feedback can help identify design issues, improve usability, and ultimately lead to a more polished and user-centered product.

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Future teams may also explore emerging topics such as artificial intelligence, fintech innovation, financial literacy, and personalized financial advisory systems. These areas offer many opportunities to create more advanced and impactful solutions.

Lastly, we strongly recommend starting the project as early as possible, maintaining regular progress reviews, and conducting user testing throughout the development process. Early feedback can help identify design issues, improve usability, and ultimately lead to a more polished and user-centered product